

GEMASTIK XIX 2026
DIVISI KOTA CERDAS

BSWMS LOOP

Bogor Smart Waste Management System - Localized Orchestration & Optimization Platform

*Product Requirements Document lengkap untuk Web, Mobile, AI-RAG, TPS 3R, Bank
Sampah, Marketplace, Pembayaran, dan Command Center*

Localized Orchestration & Optimization Platform

Product Requirements Document
Versi 1.0 - Konsep untuk Pengembangan MVP

Disusun untuk Nama Tim Gemastik
Universitas Indonesia
2026

KONTROL DOKUMEN

Atribut	Nilai
Judul	BSWMS LOOP Product Requirements Document
Versi	1.0
Status	Draft baseline untuk desain dan pengembangan MVP
Pemilik Produk	Nama Tim Gemastik / calon Product Owner DLH Kota Bogor
Perguruan Tinggi	Universitas Indonesia
Tanggal	24 Juli 2026
Klasifikasi	Publik - konsep; tanpa data internal atau rahasia

RIWAYAT REVISI

Versi	Tanggal	Penyusun	Perubahan
1.0	24 Juli 2026	Nama Tim Gemastik	Baseline PRD lengkap untuk BSWMS LOOP

PENGGUNA DOKUMEN

Peran	Penggunaan
Product Owner	Menetapkan prioritas, scope, dan acceptance
Business Analyst	Memvalidasi proses dan requirement
UX/UI Designer	Menyusun journey, wireframe, dan design system
Engineering	Membangun frontend, backend, data, integrasi, dan AI
QA/Security	Menguji fungsi, keamanan, performa, dan AI
Stakeholder Kota	Memvalidasi proses, data, kelembagaan, dan dampak

1. EXECUTIVE SUMMARY

BSWMS LOOP adalah platform pengelolaan sampah dan ekonomi sirkular Kota Bogor yang menghubungkan operasi pemerintah, TPS 3R, bank sampah, masyarakat, petugas/armada, pembeli, pengolah material, dan penyedia pembayaran. Produk dirancang untuk web dan mobile serta memiliki AI-RAG untuk analytics copilot dan chatbot. PRD ini

mendefinisikan target pengguna, scope, user journey, requirement fungsional, data, arsitektur, AI governance, pembayaran, keamanan, non-functional requirements, roadmap, dan acceptance criteria.

North-star product statement

Satu platform yang mengubah aliran sampah menjadi aliran data, material, nilai ekonomi, dan keputusan yang dapat ditelusuri.

2. PRODUCT CONTEXT

2.1 Problem

Pengelolaan sampah Kota Bogor melibatkan banyak aktor dan sistem pencatatan. Fasilitas 3R dan bank sampah membutuhkan akses pasar, pembeli membutuhkan pasokan yang konsisten, pemerintah membutuhkan data dan KPI, petugas membutuhkan workflow, dan warga membutuhkan kanal layanan yang sederhana. Tanpa platform terpadu, data dan material tidak bergerak dalam satu loop.

2.2 Product Vision

Menjadi operating platform kota yang memungkinkan pengelolaan sampah end-to-end, pengurangan residu, pemulihan nilai material, pelayanan publik berbasis data, dan keputusan yang didukung AI secara aman.

2.3 Goals

ID	Goal
G-01	Menyatukan data dan workflow TPS/TPS3R, bank sampah, armada, pengaduan, marketplace, dan pembayaran.
G-02	Meningkatkan sell-through material terpilah melalui demand discovery dan AI matching.
G-03	Menyediakan RAG chatbot dan analytics copilot yang grounded, bersitasi, dan sesuai peran.
G-04	Mencatat traceability material dan dampak lingkungan dari sumber sampah sampai pembeli/pengolah.
G-05	Menyediakan MVP yang dapat didemokan dan dipilotkan dengan integrasi bertahap.

2.4 Non-Goals

- Tidak menjadi e-wallet, bank, atau penyelenggara jasa pembayaran.
- Tidak memproses limbah B3 pada marketplace umum.
- Tidak menggantikan keputusan resmi pejabat atau operator dengan AI otonom.
- Tidak mencakup implementasi sensor seluruh kota pada release MVP.
- Tidak membangun marketplace nasional pada release pertama.
- Tidak membuka data individu atau transaksi detail sebagai open data.

3. PRODUCT PRINCIPLES

ID	Prinsip	Implikasi
P-01	Public-service first	Prioritaskan outcome layanan dan dampak, bukan engagement semata.
P-02	One input, many uses	Data dicatat sekali lalu digunakan untuk operasi, pembayaran, analitik, dan pelaporan.
P-03	Human accountable	AI memberi dukungan; manusia tetap bertanggung jawab.
P-04	Evidence by default	Jawaban AI, transaksi, dan tindakan penting memiliki sumber/bukti.
P-05	Inclusive by design	Antarmuka sederhana, assisted mode, low bandwidth, dan aksesibilitas.
P-06	Modular and interoperable	API-first, event-ready, dan dapat terhubung sistem pemerintah/mitra.
P-07	Privacy and security by design	Minimisasi data, least privilege, encryption, dan audit.

4. SCOPE DAN RELEASE STRATEGY

4.1 Scope MVP

Epic	MVP Scope
Identity & Organization	Role, user, organization, facility, verification
LOOP 3R	Penerimaan, timbang, batch, proses, stok, output, residu
Bank Sampah	Nasabah, setoran, tabungan, penarikan/payout record

Epic	MVP Scope
LOOP Market	Listing, search, RFQ/order, payment, pickup, QC, dispute
Citizen	Lokasi, jadwal, edukasi, pengaduan, setor dan status
Operations	Work order, facility status, dashboard dan basic GIS
LOOP AI	RAG chatbot, analytics copilot, document admin dan evaluation
Reporting	KPI, transaction report, diversion dan audit export

4.2 Post-MVP

Release	Fokus
R2	IoT fill-level, smart weighing integration, GPS live tracking, optimized routing
R3	Computer vision, dynamic pricing signals, advanced forecast, digital twin
R4	Inter-city configuration, open API ecosystem, carbon/ESG verification

5. USERS, PERSONAS, DAN JOBS-TO-BE-DONE

Persona	Organisasi	JTBD	Pain/Need
P1 - Operator Command Center	DLH	Memantau kondisi kota, prioritas, SLA, dan mengalokasikan tugas.	Data real-time, alert yang dapat dipercaya, peta, tindakan cepat.
P2 - Pengelola TPS3R	Komunitas/UPTD	Mencatat material, proses, stok, menjual output, dan meminta pickup.	Input sederhana, stok akurat, pembeli, pembayaran.
P3 - Pengelola Bank Sampah	Komunitas	Mengelola nasabah, setoran, tabungan, batch, dan penjualan.	Transparansi saldo, harga, laporan, buyer matching.
P4 - Petugas/Driver	Lapangan	Menyelesaikan work order dan membuktikan layanan.	Rute, tugas jelas, offline, scan, foto, geotag.
P5 - Warga	Masyarakat	Mengetahui cara memilah, titik setor, jadwal, dan melapor.	Bahasa sederhana, lokasi, chatbot, status transparan.
P6 - Pembeli/Pengolah	Industri/UMKM	Mencari pasokan sesuai spesifikasi dan menyelesaikan pembelian.	Kualitas, kuantitas, harga, pickup, invoice.

Persona	Organisasi	JTBD	Pain/Need
P7 - Pimpinan/Analisis	Pemkot	Memahami kinerja, risiko, dan dampak untuk kebijakan.	KPI, tren, penjelasan AI, sumber, export.
P8 - Administrator Knowledge	DLH/Kominfo	Mengelola sumber RAG dan akses.	Versioning, approval, metadata, audit.

5.1 Role Matrix

Role	Scope	Hak Utama
Citizen	Publik/terdaftar	View public info, complaint, deposit request, marketplace buy (opsional)
Facility Operator	TPS3R/Bank Sampah	Batch, processing, stock, listing, order fulfillment
Collector/Driver	Operator logistik	Pickup jobs, route, proof
Buyer	Verified organization	Search, order, payment, QC, dispute
Command Center	DLH	Monitor, dispatch, override, incident, analytics
Executive	Pemkot	Read-only KPI, copilot, reports
Knowledge Admin	DLH/Kominfo	Source ingestion, approval, access policy
System Admin	Kominfo	Tenant, config, security, integration, audit

6. END-TO-END USER JOURNEYS

Journey	Actors	Flow
J-01 Material sale	Operator -> Buyer	Receive -> batch -> process -> stock -> listing -> match -> order -> pay -> pickup -> QC -> settle -> impact
J-02 Citizen deposit	Citizen -> Bank Sampah	Find facility -> check accepted materials -> schedule/drop-off -> weigh -> receipt -> balance/history
J-03 Complaint	Citizen -> Command Center	Submit photo/location -> classify -> verify -> work order -> resolve -> feedback
J-04 AI analytics	Executive/Operator	Ask question -> retrieve data/docs -> compute -> cite -> recommend -> create action (confirmed)

Journey	Actors	Flow
J-05 Knowledge update	Admin	Upload -> classify -> scan -> chunk -> approve -> index -> evaluate -> publish

7. INFORMATION ARCHITECTURE

Navigation	Sub-domain
Dashboard	Role-based summary, alerts, tasks, KPI
Peta	Facilities, complaints, pickup, service areas, demand/supply
Operasi	Work order, incidents, ritasi/pickup, SLA
Fasilitas 3R	Profile, intake, processing, stock, equipment, reports
Bank Sampah	Members, deposits, balance, batch, payout records
Marketplace	Catalog, listing, RFQ, orders, payments, disputes
AI Copilot	Conversation, sources, charts, action suggestions
Knowledge	Sources, versions, approvals, evaluation
Reports	Operational, financial, environment, audit
Admin	Users, roles, master data, integrations, policies

8. FUNCTIONAL REQUIREMENTS

Requirement notation

Priority: Must / Should / Could. Acceptance criteria ditulis agar dapat diuji. Semua event kritis harus memiliki actor, timestamp, correlation ID, dan audit log.

8.1 Identity, Organization, and Access

ID	Priority	Requirement	Acceptance Criteria
IAM-001	Must	Sistem mendukung login email/nomor ponsel dan opsi SSO pemerintah.	Pengguna valid dapat login; sesi dan refresh token dikelola aman.
IAM-002	Must	Mendukung MFA untuk admin, command center, knowledge admin, dan finance role.	Login role sensitif meminta faktor kedua.

ID	Priority	Requirement	Acceptance Criteria
IAM-003	Must	RBAC dengan scope organisasi/fasilitas/wilayah.	Pengguna tidak dapat membaca/ubah data di luar scope.
IAM-004	Must	Onboarding organisasi dan verifikasi dokumen.	Status Draft, Submitted, Verified, Rejected tercatat.
IAM-005	Must	Delegasi operator per fasilitas.	Owner dapat menambah/mencabut operator; perubahan diaudit.
IAM-006	Should	ABAC untuk data knowledge sensitif berdasarkan classification.	Retriever hanya mengembalikan chunk yang diizinkan.
IAM-007	Must	Consent dan privacy notice untuk pengguna publik.	Versi consent dan timestamp tersimpan.
IAM-008	Must	Account recovery dan session revocation.	Admin/pengguna dapat mencabut seluruh sesi.

8.2 Master Data and Facility Registry

ID	Priority	Requirement	Acceptance Criteria
MD-001	Must	CRUD wilayah, fasilitas, tipe fasilitas, service area, jam operasi.	Data tervalidasi dan dapat dipetakan.
MD-002	Must	Katalog material: kategori, unit, grade, contamination, handling.	Katalog versioned dan dipakai lintas modul.
MD-003	Must	Profil TPS3R/bank sampah: kapasitas, equipment, accepted materials.	Profil tampil pada portal sesuai visibilitas.
MD-004	Should	Import bulk CSV dengan preview dan error report.	Baris invalid tidak masuk dan dapat diunduh.
MD-005	Must	Audit perubahan master data.	Before/after dan actor tersedia.
MD-006	Should	Data stewardship assignment.	Setiap domain memiliki owner dan steward.

8.3 TPS 3R Operations

ID	Priority	Requirement	Acceptance Criteria
T3R-001	Must	Mencatat intake material dengan batch ID unik.	Batch ID tidak duplikat; berat, jenis, sumber wajib.
T3R-002	Must	Mendukung bukti timbang foto/device reading.	Bukti terhubung batch dan tidak dapat diganti tanpa audit.

ID	Priority	Requirement	Acceptance Criteria
T3R-003	Must	Mencatat processing run: input, metode, output, residu, durasi.	Mass balance dihitung dan flag jika di luar toleransi.
T3R-004	Must	Stock ledger per material/grade/location.	Setiap movement menghasilkan ledger immutable.
T3R-005	Should	Alert umur stok, kapasitas, dan kebutuhan maintenance.	Alert muncul berdasarkan threshold konfigurabel.
T3R-006	Must	Membuat listing dari stock yang eligible.	Kuantitas listing tidak melebihi available stock.
T3R-007	Should	Dashboard yield, cost, diversion, and revenue.	Filter per periode dan export.
T3R-008	Could	Integrasi sensor lingkungan/peralatan.	Telemetry tampil dengan health status.

8.4 Bank Sampah Operations

ID	Priority	Requirement	Acceptance Criteria
BS-001	Must	Registrasi nasabah dan household/organization account.	ID unik dan minimisasi data pribadi.
BS-002	Must	Setoran multi-material dengan timbang dan harga snapshot.	Receipt menunjukkan berat, harga, nilai, operator.
BS-003	Must	Ledger tabungan immutable.	Saldo dihitung dari ledger, bukan field editable.
BS-004	Must	Penggabungan setoran menjadi batch penjualan.	Traceability nasabah ke batch tersimpan secara terbatas.
BS-005	Must	Penarikan/payout record dengan approval.	Status dan bukti payout tercatat.
BS-006	Should	Statement nasabah dan laporan periodik.	PDF/CSV tersedia sesuai role.
BS-007	Should	Price board per facility dan effective date.	Harga historis tidak berubah retroaktif.
BS-008	Could	Gamification komunitas tanpa dark pattern.	Reward terkait aksi nyata dan dapat diaudit.

8.5 Marketplace Listings and Discovery

ID	Priority	Requirement	Acceptance Criteria
MKT-001	Must	Create listing dari verified stock.	Material, grade, qty, unit, price, location, expiry wajib.
MKT-002	Must	Search/filter kategori, jarak, qty, grade, price, seller.	Hasil konsisten dan paginated.

ID	Priority	Requirement	Acceptance Criteria
MKT-003	Should	Map view supply-demand.	Marker di-cluster dan privacy location diterapkan.
MKT-004	Must	Seller verification badge dan facility profile.	Badge hanya muncul setelah verification.
MKT-005	Must	Listing lifecycle Draft/Published/Reserved/Sold/Expired/Cancelled.	Transition mengikuti state machine.
MKT-006	Should	RFQ untuk kebutuhan berulang/volume besar.	Buyer dapat menetapkan spec dan due date.
MKT-007	Must	Prevent overselling melalui reservation.	Concurrent order tidak menjual stock melebihi available.
MKT-008	Should	AI match score dengan explanation.	Faktor match ditampilkan tanpa mengungkap data sensitif.

8.6 Orders, Payments, Settlement, and Disputes

ID	Priority	Requirement	Acceptance Criteria
PAY-001	Must	Order dengan item, qty, price, fees, taxes (jika berlaku), delivery.	Total dihitung server-side.
PAY-002	Must	Payment initiation melalui PJP berizin.	User diarahkan/SDK PJP; platform tidak menyimpan credential pembayaran.
PAY-003	Must	Webhook signature verification dan idempotency.	Duplicate callback tidak menduplikasi pembayaran.
PAY-004	Must	Payment state Pending/Paid/Failed/Expired/Refunded/Partially Refunded.	State selaras dengan provider dan ledger.
PAY-005	Must	Settlement hanya setelah delivery/QC acceptance.	Settlement event memiliki evidence dan approval.
PAY-006	Must	Reconciliation report antara order, payment, fee, settlement.	Mismatch terdeteksi dan masuk exception queue.
PAY-007	Must	Dispute workflow untuk berat, kualitas, keterlambatan.	Evidence upload, response deadline, mediator, resolution.
PAY-008	Should	Partial adjustment jika berat aktual berbeda.	Adjustment memerlukan acceptance dan tercatat.
PAY-009	Must	Invoice/receipt downloadable.	Dokumen memiliki order ID dan status.
PAY-010	Should	Refund via provider sesuai capability.	Refund result di-sync dan diaudit.

8.7 Pickup, Logistics, and Work Orders

ID	Priority	Requirement	Acceptance Criteria
LOG-001	Must	Create pickup job dari order atau work order.	Job memiliki origin, destination, material, window.
LOG-002	Must	Assignment ke driver/vehicle/operator.	Assignment mempertimbangkan scope dan availability.
LOG-003	Must	Status Assigned/Accepted/En Route/Arrived/Loaded/Delivered/Closed.	Timestamp dan geotag per milestone.
LOG-004	Must	Proof of pickup/delivery: photo, signature/OTP, weight.	Evidence required sebelum close.
LOG-005	Should	Route suggestion berbasis jarak dan constraint.	Route dapat di-override dengan alasan.
LOG-006	Should	Offline task cache dan sync.	Petugas dapat bekerja saat koneksi putus.
LOG-007	Could	GPS live tracking and route compliance.	Consent dan retention diterapkan.

8.8 Citizen Services and Complaints

ID	Priority	Requirement	Acceptance Criteria
CIT-001	Must	Public directory fasilitas dan accepted materials.	Search by location dan jam operasi.
CIT-002	Must	Jadwal layanan/pickup area.	Pengguna melihat status perubahan jadwal.
CIT-003	Must	Complaint dengan kategori, deskripsi, foto, lokasi.	Spam/rate limit dan privacy masking diterapkan.
CIT-004	Must	Complaint status dan notification.	Citizen menerima acknowledgement dan resolution.
CIT-005	Should	Deposit appointment/request pickup.	Facility dapat accept/reschedule.
CIT-006	Must	Education content dan RAG chatbot.	Jawaban mencantumkan sumber/abstain.
CIT-007	Should	Feedback dan satisfaction rating.	Tidak dapat rating sebelum case closed.
CIT-008	Should	Assisted mode untuk operator kelurahan/call center.	Aksi atas nama warga memiliki consent marker.

8.9 Command Center and Analytics

ID	Priority	Requirement	Acceptance Criteria
OPS-001	Must	Role-based dashboard KPI dan alerts.	KPI definisi dan timestamp data terlihat.
OPS-002	Must	Map layers: facilities, complaints, pickup, supply-demand.	Layer toggle dan filter wilayah.
OPS-003	Must	Alert triage dan assign action.	Alert status/owner/SLA tercatat.
OPS-004	Should	Threshold configuration per area/facility.	Perubahan threshold versioned.
OPS-005	Must	Export report CSV/PDF.	Export mengikuti access scope.
OPS-006	Should	Scenario comparison dan trend.	Metode/periode ditampilkan.
OPS-007	Must	Data quality dashboard.	Missing, stale, duplicate, anomaly indicators.
OPS-008	Should	Public aggregate dashboard.	Tidak mengandung PII dan small-cell disclosure.

8.10 AI-RAG Chatbot and Copilot

ID	Priority	Requirement	Acceptance Criteria
AI-001	Must	RAG answer dari approved knowledge sources.	Jawaban faktual memiliki citation atau abstain.
AI-002	Must	Role-aware retrieval.	Chunk di luar clearance tidak diretrieval.
AI-003	Must	Hybrid retrieval + metadata filters + reranking.	Evaluation menunjukkan recall target.
AI-004	Must	Conversation context terbatas dan dapat dihapus.	Retention dan privacy setting diterapkan.
AI-005	Must	Tool calling untuk query data/analytics terverifikasi.	LLM tidak mengarang nilai KPI.
AI-006	Must	Action suggestion memerlukan user confirmation.	Tidak ada autonomous financial/operational execution.
AI-007	Must	Prompt injection detection dan untrusted content isolation.	Dokumen dengan instruksi jahat tidak mengubah system policy.
AI-008	Must	Feedback thumbs up/down + reason.	Feedback masuk evaluation backlog.
AI-009	Must	Response metadata: source, date, confidence/coverage, limitations.	Metadata tampil di UI.

ID	Priority	Requirement	Acceptance Criteria
AI-010	Should	Model fallback dan circuit breaker.	Jika provider gagal, user mendapat fallback/clear error.
AI-011	Should	Cost and token budget control.	Quota per role dan monitoring tersedia.
AI-012	Must	AI audit log dengan prompt hash, model, sources, latency.	Log aman dan akses terbatas.

8.11 Knowledge Management

ID	Priority	Requirement	Acceptance Criteria
KN-001	Must	Upload PDF/DOCX/HTML/text dengan metadata.	Type, owner, classification, effective date wajib.
KN-002	Must	Approval workflow Draft/Review/Approved/Archived.	Hanya approved masuk production index.
KN-003	Must	Versioning dan superseded relationship.	Retriever memprioritaskan current effective version.
KN-004	Must	Chunk preview dan redaction sebelum indexing.	Admin dapat mengecualikan halaman/chunk.
KN-005	Should	Automated quality checks: empty, duplicate, OCR quality.	Issue ditampilkan sebelum approve.
KN-006	Must	Delete/unpublish propagates ke vector index.	Dokumen tidak muncul setelah SLA deletion.
KN-007	Should	Evaluation set management.	Question, expected evidence, expected answer tersimpan.

8.12 Notifications and Communication

ID	Priority	Requirement	Acceptance Criteria
NOT-001	Must	In-app notification untuk events penting.	Unread count dan deep link.
NOT-002	Should	Email/WhatsApp/SMS provider abstraction.	Channel sesuai consent dan priority.
NOT-003	Must	Template versioning dan localization.	Template change diaudit.
NOT-004	Must	Delivery status dan retry.	Failed messages masuk dead-letter queue.
NOT-005	Should	Quiet hours dan user preference.	Emergency override terbatas.

8.13 Administration and Audit

ID	Priority	Requirement	Acceptance Criteria
ADM-001	Must	System configuration tanpa redeploy untuk thresholds, fees, categories.	Change memiliki effective date.
ADM-002	Must	Immutable audit log untuk critical actions.	Search by actor, entity, date, correlation ID.
ADM-003	Must	Feature flags per pilot area.	Release dapat dibatasi.
ADM-004	Must	Integration credential vault and rotation.	Secret tidak tampil plaintext.
ADM-005	Should	Support console read-only impersonation with consent.	Session direkam dan dibatasi.
ADM-006	Must	Data retention and deletion job.	Policy per data class dapat dikonfigurasi.

9. BUSINESS RULES

ID	Rule
BR-001	Available stock = inbound + adjustments - reserved - sold - waste/loss.
BR-002	Listing hanya dapat dibuat dari stock milik organisasi/fasilitas yang verified.
BR-003	Kuantitas order final menggunakan berat aktual jika disepakati dan tidak melebihi toleransi.
BR-004	Settlement tidak dapat dipicu sebelum Paid + Delivered + QC Accepted.
BR-005	Refund dan adjustment harus mereferensikan payment/order original.
BR-006	Harga historis setoran bank sampah tidak boleh berubah setelah receipt final.
BR-007	Dokumen knowledge Archived/Superseded tidak digunakan untuk jawaban baru kecuali konteks historis eksplisit.
BR-008	AI harus abstain jika evidence coverage di bawah threshold atau sumber saling konflik tanpa resolusi.
BR-009	Data publik tidak menampilkan detail transaksi individu, saldo, alamat presisi rumah, atau identifier pribadi.
BR-010	Material yang termasuk B3 ditolak atau diarahkan ke kanal khusus berizin.

10. STATE MACHINES

10.1 Listing State

Current	Action	Next	Guard
Draft	Publish	Published	Seller verified; stock available; required fields complete
Published	Reserve	Reserved	Atomic stock reservation
Reserved	Payment expired/cancel	Published	Release reservation
Reserved	Paid	Sold	Payment verified
Published	Expire	Expired	Expiry reached
Any non-final	Cancel	Cancelled	No settled order

10.2 Order and Payment State

State	Allowed Next	Notes
Created	Awaiting Payment, Cancelled	Invoice generated
Awaiting Payment	Paid, Expired, Failed	Provider status
Paid	Pickup Scheduled, Refunded	Stock locked
Pickup Scheduled	In Transit, Cancelled by mediation	Job assigned
In Transit	Delivered	Proof required
Delivered	QC Accepted, Disputed	Buyer window applies
QC Accepted	Settled	Settlement instruction
Disputed	Adjusted, Refunded, QC Accepted	Mediator decision
Settled	Closed	Final

11. DATA MODEL

Entity	Key Fields
User	user_id, identity, contact, status, consent
Organization	org_id, type, verification, legal profile
Facility	facility_id, org_id, type, geo, capacity, services
MaterialCatalog	material_id, category, unit, grade schema
Batch	batch_id, source, facility, material, weight, quality, status

Entity	Key Fields
ProcessingRun	run_id, method, input batches, outputs, residue
StockLedger	entry_id, facility, material, qty, movement_type, reference
BankMember	member_id, facility, profile_minimal, consent
SavingsLedger	entry_id, member, deposit/payout, amount, reference
Listing	listing_id, stock_ref, price, qty, expiry, state
Order	order_id, buyer, seller, items, amounts, status
Payment	payment_id, provider_ref, method, amount, status
Settlement	settlement_id, order, beneficiary, amount, status
PickupJob	job_id, origin, destination, assignment, evidence
Complaint	complaint_id, reporter, location, category, SLA, status
KnowledgeDocument	doc_id, version, owner, classification, effective_date
KnowledgeChunk	chunk_id, doc_id, text, embedding_ref, ACL, metadata
AIInteraction	interaction_id, user, model, sources, latency, feedback
AuditEvent	event_id, actor, action, entity, before_hash, after_hash
ImpactLedger	impact_id, order/batch, diverted_weight, method, calculation_version

11.1 Data Classification

Class	Examples	Control
Public	Aggregate dashboard, facility public profile, education	Open/portal
Internal	Operational KPI, non-sensitive facility data	Authenticated government/facility
Confidential	Transaction detail, contract, organizational finance	Need-to-know
Personal	Name, phone, address, bank/payment identifiers	Consent + strict RBAC
Restricted	Security config, credentials, sensitive incident	Security/admin only

12. API AND INTEGRATION REQUIREMENTS

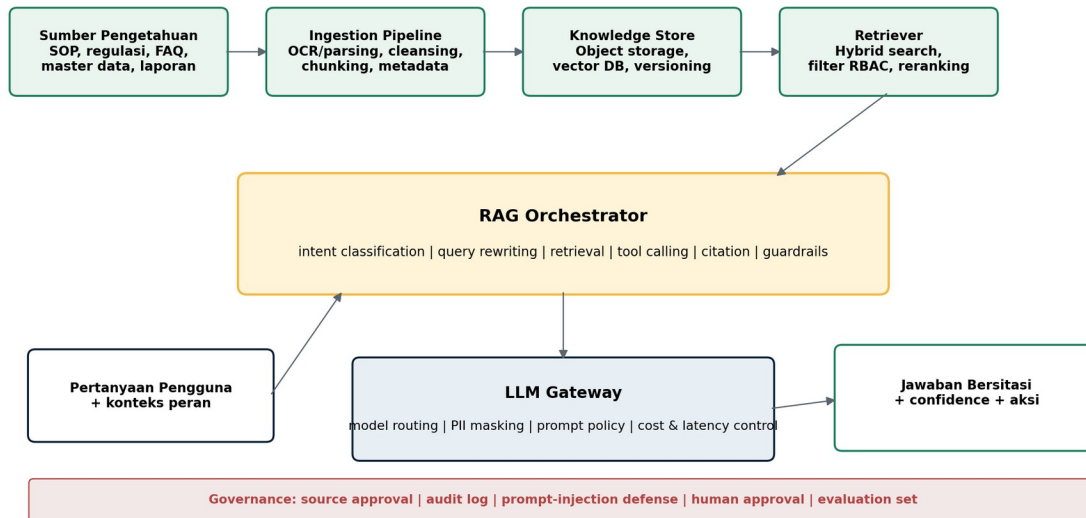
ID	Integration	Requirement
API-01	REST/JSON API with OpenAPI spec	Versioned /v1, error schema, correlation ID
API-02	Payment provider	Create payment, webhook, status, refund, settlement/reconciliation
API-03	Maps/GIS	Geocoding, routing, tiles; caching and quota control
API-04	Notification provider	Email/WhatsApp/SMS abstraction and delivery callback
API-05	IoT/GPS future	MQTT/HTTP ingestion, device identity, timestamp validation
API-06	Government integration	SSO, Satu Data export, complaint or command center gateway
API-07	AI provider	Model gateway, streaming, timeout, safety, cost metadata

12.1 API Standards

- OAuth 2.0/OIDC for user/service authentication; mTLS or signed request for sensitive machine integration.
- Idempotency-Key for create payment, order, settlement, and intake operations.
- Webhook signature, timestamp tolerance, replay protection, and delivery retry.
- Pagination, filter, sort, field selection, and consistent UTC timestamps with local display.
- PII masking in logs; request/response body logging disabled for sensitive endpoints.
- Rate limiting per identity, IP, organization, and endpoint risk.

13. AI-RAG PRODUCT SPECIFICATION

Arsitektur AI-RAG yang Terverifikasi dan Berbasis Peran



Arsitektur logical AI-RAG BSWMS LOOP.

13.1 Knowledge Sources

Source Type	Examples	Access
Public regulation	UU, Perpres, Permen, Perda, policy	Public
Operational SOP	Collection, 3R, complaint, safety, marketplace	Internal/Confidential
Facility data	Profile, capacity, accepted materials, schedule	Public/Internal
Analytics semantic layer	KPI definitions, metric queries, dimensions	Internal
Marketplace data	Listing/order aggregate, quality, demand	Role-scoped
Education/FAQ	Sorting, material handling, citizen service	Public

13.2 RAG Pipeline

Stage	Flow
Ingest	Fetch/upload -> malware scan -> parse/OCR -> normalize

Stage	Flow
Govern	Classify -> owner -> effective date -> review -> approval
Index	Chunk -> metadata -> embedding -> vector + keyword index
Query	Intent -> rewrite -> access filter -> retrieve -> rerank
Generate	Prompt with evidence -> tool calls -> answer -> citation
Validate	Grounding checks -> policy -> PII -> abstention
Observe	Log -> feedback -> eval -> issue triage -> improve

13.3 AI Use Cases and Tools

Use Case	Mechanism	Control
Citizen FAQ	RAG only	Facility directory, SOP, education
KPI explanation	SQL/metric tool + RAG	Verified numbers + definition/source
Overflow forecast	Forecast service + RAG	Prediction + factors + recommended action
Material matching	Matching service	Score and explanation
Report drafting	Data query + RAG	Draft with citations; user review
Work order creation	Suggestion + confirmation	No autonomous execution

13.4 AI Evaluation

Metric	Target MVP	Definition
Retrieval Recall@K	≥ 0.90 on curated set	Relevant evidence found
Citation Precision	≥ 0.90	Citation supports claim
Groundedness	≥ 0.90	Answer entailed by evidence/tool
Answer Relevance	≥ 0.85	Addresses user intent
Abstention Correctness	≥ 0.90	Refuses unsupported question
Sensitive Data Leakage	0 critical cases	No unauthorized disclosure
Prompt Injection Success	$< 1\%$ in red-team suite	System policy preserved
P95 Latency	≤ 8 s chatbot; ≤ 15 s analytics	Usable response time
Cost per interaction	Budget by role/use case	Operational control

13.5 AI Failure Behavior

- Jika sumber tidak cukup: jawab bahwa informasi belum tersedia dan tampilkan sumber yang dicari.
- Jika sumber konflik: tampilkan konflik, tanggal, dan minta pengguna memilih konteks atau eskalasi.
- Jika tool data gagal: jangan menebak; tampilkan status dan retry option.
- Jika pertanyaan di luar hak akses: jelaskan bahwa akses tidak tersedia tanpa membocorkan eksistensi konten sensitif.
- Jika pengguna meminta aksi: tampilkan preview dan confirmation; rekam siapa yang mengonfirmasi.

14. SECURITY, PRIVACY, AND COMPLIANCE

Domain	Requirement
Authentication	OIDC/OAuth2, MFA, strong session management
Authorization	RBAC + organization/facility/region scopes; ABAC for knowledge
Data protection	TLS, encryption at rest, key rotation, encrypted backups
Application security	SAST, DAST, dependency scan, secret scan, OWASP ASVS
API security	Rate limit, schema validation, idempotency, webhook verification
LLM security	Prompt isolation, allowlisted tools, output validation, content filtering, RAG ACL
Payment security	No card data storage, licensed provider, reconciliation, antifraud signals
Privacy	Consent, minimization, retention, access/correction/deletion workflow
Audit	Immutable event log, time sync, alert on privileged actions
Incident response	Severity, escalation, containment, notification, postmortem

14.1 Threat Scenarios

ID	Threat	Control
T-01	Account takeover	MFA, risk login, session revoke, alert

ID	Threat	Control
T-02	Cross-tenant data access	Scoped queries, policy tests, object authorization
T-03	Payment webhook spoofing	Signature, timestamp, IP/provider checks, idempotency
T-04	Stock manipulation	Immutable ledger, dual approval for adjustment
T-05	Prompt injection in document	Sanitize, isolate content, no instruction authority from retrieved text
T-06	LLM unauthorized tool call	Allowlist, schema validation, explicit confirmation
T-07	PII in logs/vector store	Redaction, classification, restricted indexing
T-08	Fraudulent quality evidence	Timestamp/geotag, device metadata, review sampling

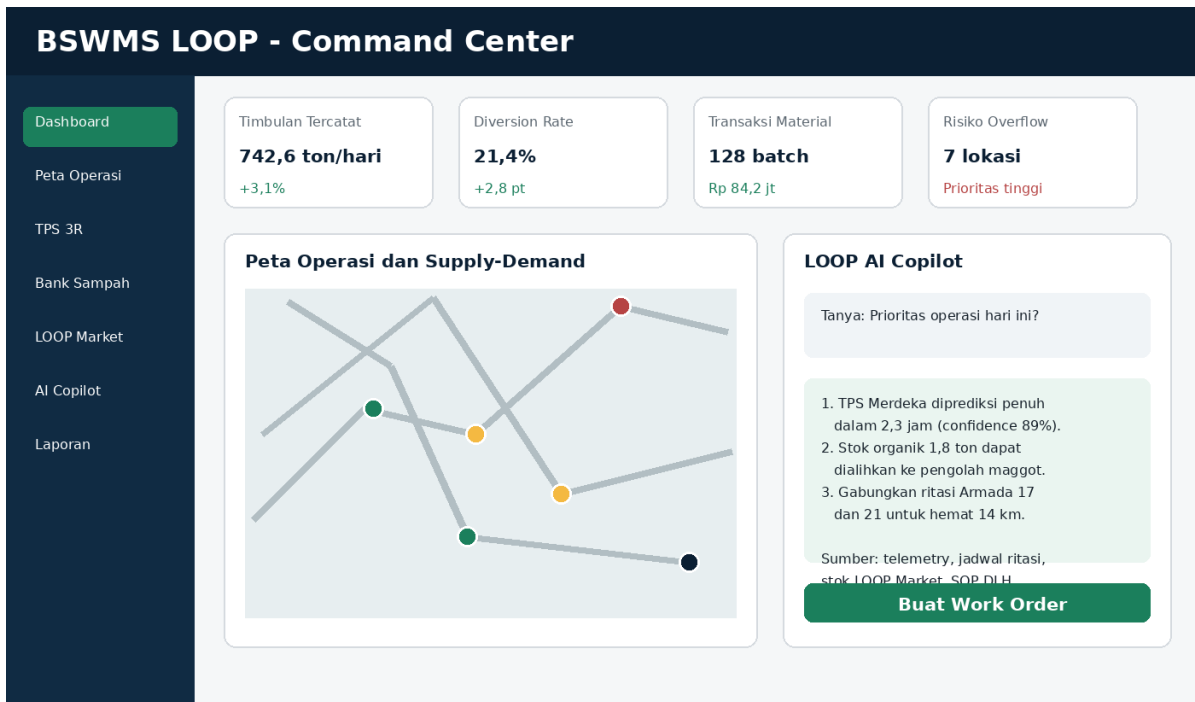
15. NON-FUNCTIONAL REQUIREMENTS

ID	Area	Target
NFR-001	Availability	MVP 99.5% monthly excluding planned maintenance
NFR-002	Performance	P95 API read < 700 ms; write < 1.2 s excluding external services
NFR-003	Marketplace search	P95 < 1.5 s for pilot dataset
NFR-004	Scalability	Horizontal scaling for stateless services; partition-ready ledgers
NFR-005	Reliability	Exactly-once business effect through idempotency; at-least-once event delivery
NFR-006	Backup	Daily full + point-in-time where available; restore test quarterly
NFR-007	RPO/RTO	MVP RPO ≤ 24 h, RTO ≤ 8 h; payment ledger stricter per provider design
NFR-008	Accessibility	WCAG 2.1 AA target for core flows
NFR-009	Low bandwidth	Core mobile screens usable on unstable 3G; image compression and retry
NFR-010	Offline	Pickup and intake draft offline queue with conflict handling

ID	Area	Target
NFR-011	Localization	Bahasa Indonesia default; content ready for local terminology
NFR-012	Observability	Metrics, logs, traces, business events and alerting
NFR-013	Maintainability	Automated tests, code standards, API docs, runbook
NFR-014	Compatibility	Latest two major Chrome/Edge/Firefox; supported Android baseline

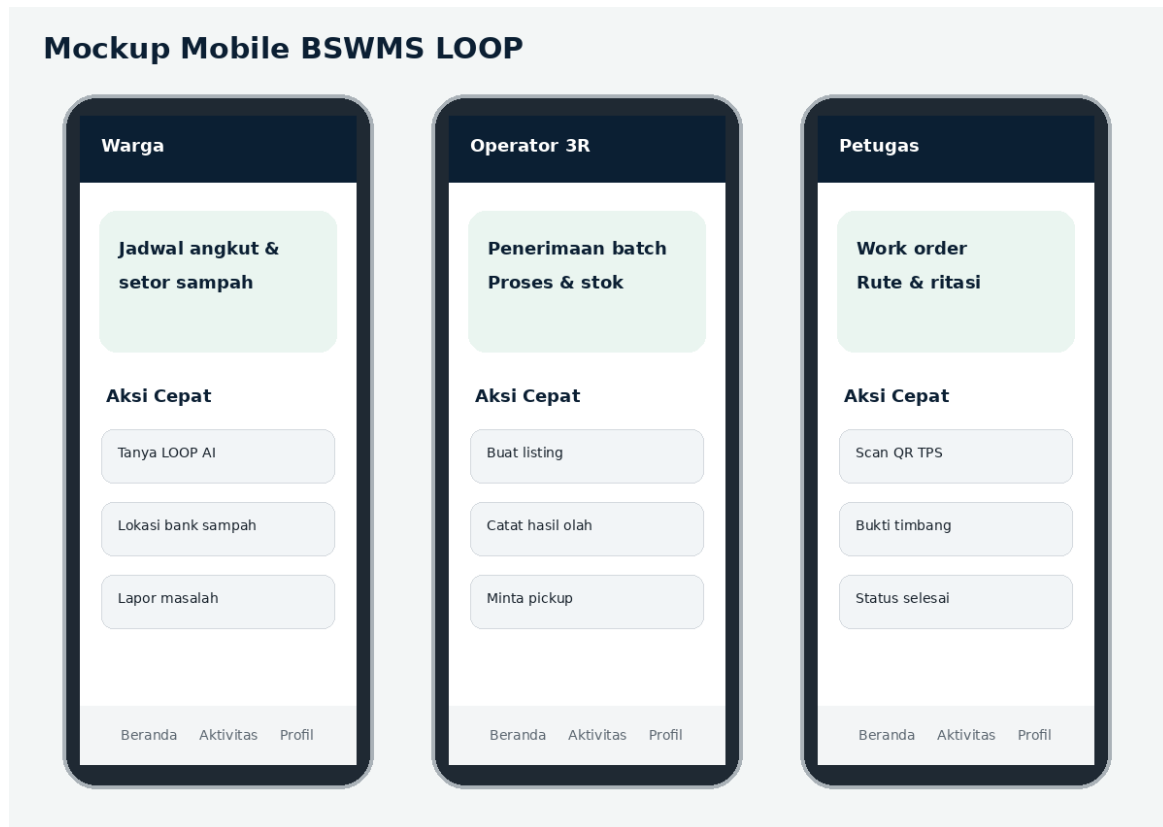
16. UX, ACCESSIBILITY, AND DESIGN SYSTEM

Principle	Requirement
Simple first	Primary task max 3-5 major steps; progressive disclosure
Evidence visible	Status, source, timestamp, and proof near decision
Error recovery	Draft autosave, clear retry, no silent failure
Role-based home	Dashboard differs for citizen, facility, buyer, operator
Accessibility	Keyboard, contrast, labels, screen reader, text alternatives
Trust	No dark patterns; fees, AI limitations, and payment status explicit
Field usability	Large touch targets, scan-first, offline, photo compression



Reference mockup: Command Center dan AI Copilot.





Reference mockup: aplikasi mobile multi-role.

17. PRODUCT ANALYTICS AND KPI

Domain	Metrics
Acquisition	Verified facilities, buyers, citizens registered
Activation	First batch, first listing, first order, first AI grounded answer
Engagement	WAU by role, tasks completed, query success
Marketplace	GMV, sell-through, time-to-match, order completion, dispute rate
Operations	SLA, work order completion, pickup punctuality, data freshness
Environment	Diverted weight, residue, processing yield, estimated emissions avoided
AI	Groundedness, citation precision, abstention, feedback, latency, cost
Quality	Missing data, duplicate rate, reconciliation mismatch

17.1 Event Taxonomy

Event	Properties (non-exhaustive)
user_logged_in	role, org, channel
batch_created	facility, material, weight, source
processing_completed	method, input, output, residue
listing_published	material, qty, price, seller
match_viewed	listing, buyer, score_band
order_created	buyer, seller, value
payment_status_changed	provider, status, value
pickup_completed	distance, duration, actual_weight
complaint_resolved	category, SLA, resolution
ai_answer_generated	use_case, sources_count, grounded_score, latency
ai_feedback_submitted	rating, reason

18. TESTING AND QUALITY ASSURANCE

Test Type	Coverage
Unit	Business rules, state transitions, calculations, parsers
Integration	Payment, webhook, notification, map, AI provider
Contract	API schema and provider sandbox
E2E	Top journeys across web/mobile
Security	SAST/DAST, authz, secret, dependency, penetration test
Performance	Load, stress, soak, database query
Data	Migration, reconciliation, mass balance, quality rules
AI evaluation	Offline benchmark, adversarial prompts, role leakage, citations
Usability	Operators, bank sampah, citizens, buyers
Accessibility	Automated + manual keyboard/screen reader
DR	Backup restore and recovery drill

18.1 UAT Exit Criteria

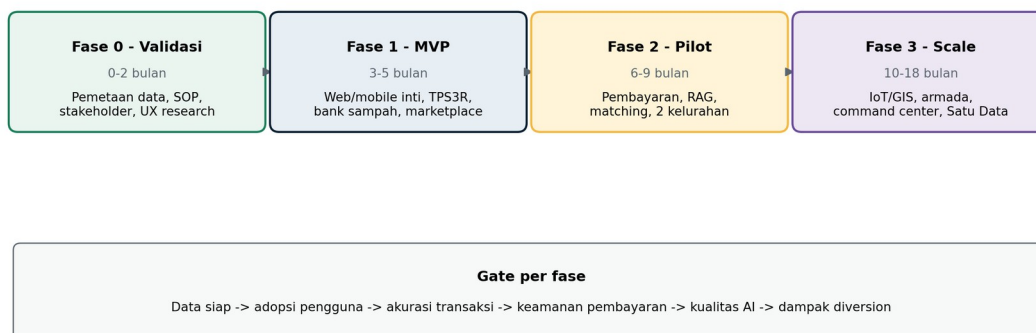
- 100% Must requirements lulus atau memiliki approved waiver.
- Tidak ada severity Critical/High terbuka pada security dan payment flow.
- Payment reconciliation sandbox 100% untuk test cases.
- Top 5 user journeys berhasil $\geq 95\%$ pada scripted UAT.
- AI groundedness dan citation precision memenuhi target MVP.
- Backup restore dan rollback release berhasil.
- Product Owner dan stakeholder pilot menandatangani go-live checklist.

19. DEPLOYMENT AND OPERATIONS

Area	Requirement
Environments	Local, Dev, Test, Staging/UAT, Production
CI/CD	Build, test, security scan, artifact signing, deploy, smoke test
Release	Feature flags, canary/pilot area, rollback
Secrets	Central vault, no secrets in repo, rotation
Database	Migration versioning, backup before migration, rollback plan
Observability	SLO dashboard, error budget, on-call alert
Runbooks	Payment outage, AI outage, data sync, incident, restore

20. ROADMAP AND DELIVERY PLAN

Roadmap Implementasi Bertahap



Roadmap produk dan implementasi.

Timebox	Theme	Deliverables
Sprint 0-2	Discovery and foundation	Process mapping, data dictionary, architecture, design system, CI/CD
Sprint 3-5	Core 3R and bank	Identity, facility, batch, stock, member/deposit
Sprint 6-8	Marketplace	Listing, search, order, payment sandbox, pickup
Sprint 9-10	Citizen and operations	Complaint, directory, dashboard, basic map
Sprint 11-12	RAG MVP	Knowledge ingestion, chatbot, citations, evaluation
Sprint 13-14	Analytics copilot	Metric tools, dashboard explanation, action preview
Sprint 15	Hardening	Security, performance, accessibility, UAT
Sprint 16	Demo/pilot readiness	Data seed, training, runbook, video and presentation

21. PRIORITIZED MVP BACKLOG

ID	Priority	Backlog Item
MVP-01	Must	Login, role, organization/facility scope
MVP-02	Must	Facility registry and material catalog
MVP-03	Must	TPS3R intake, processing, stock ledger
MVP-04	Must	Bank member, deposit, savings ledger
MVP-05	Must	Listing/search/order/reservation
MVP-06	Must	Payment sandbox + webhook + reconciliation
MVP-07	Must	Pickup proof and QC acceptance
MVP-08	Must	Command center KPI and basic map
MVP-09	Must	Citizen directory, complaint and status
MVP-10	Must	Knowledge upload, approval and indexing

ID	Priority	Backlog Item
MVP-11	Must	RAG chatbot with citations and abstention
MVP-12	Must	Analytics copilot with verified metric tools
MVP-13	Must	Audit log and security controls
MVP-14	Should	Offline queue for field intake/pickup
MVP-15	Should	AI matching and explanation
MVP-16	Should	Public aggregate impact dashboard

22. RACI

Area	A	R	C	I
Product scope	PO DLH	Tim Produk	Kominfo, Bappeda, operator	Steering
Business process	DLH	BA	TPS3R, bank sampah, driver	Tim Teknis
Architecture/security	Kominfo	Tech Lead/Security	DLH, PJP	Steering
Marketplace rules	DLH/Product	BA/Legal	Seller, buyer, PJP	Users
AI knowledge	Knowledge Owner	Knowledge Admin	Legal, data steward	Users
AI model/evaluation	AI Lead	ML/AI Team	PO, security, domain experts	Steering
Payment integration	Product/Finance	Backend/PJP	Legal, security	Seller/buyer
UAT	Product Owner	QA	All pilot roles	Steering

23. RISKS, ASSUMPTIONS, AND DEPENDENCIES

Type	Item	Impact	Treatment
Risk	Data inconsistency	High	Data profiling, mapping, steward, quality gate
Risk	Low operator adoption	High	Co-design, simple UI, training, assisted mode
Risk	Payment regulation/provider limitation	High	Licensed PJP, legal review, provider abstraction

Type	Item	Impact	Treatment
Risk	AI hallucination or leakage	High	RAG, ACL, eval, red-team, abstain, audit
Risk	Marketplace liquidity low	Medium	Anchor buyers, demand onboarding, RFQ, pilot categories
Risk	Quality dispute	High	Grade standard, evidence, tolerance, mediation
Assumption	Pilot partners provide representative data	-	Validate during discovery
Assumption	PJP sandbox available	-	Select provider early
Dependency	Maps/geocoding provider	-	Quota and fallback
Dependency	Regulatory/SOP approval	-	Governance workstream

24. DEFINITION OF READY AND DEFINITION OF DONE

24.1 Definition of Ready

- Problem, actor, value, priority, and acceptance criteria defined.
- UX flow/wireframe and data fields available.
- Dependencies and security/privacy impact identified.
- Test data and external sandbox available where needed.
- Product Owner agrees scope and out-of-scope.

24.2 Definition of Done

- Code reviewed, automated tests pass, and documentation updated.
- Acceptance criteria pass in target environment.
- Security and privacy checks pass.
- Telemetry, logs, and business events implemented.
- Accessibility and responsive behavior checked.
- Release notes, rollback, and runbook updated.
- Product Owner accepts.

25. OPEN QUESTIONS

ID	Question	Owner	Due
OQ-01	Wilayah dan jumlah fasilitas pilot pertama?	PO/DLH	Before Sprint 1
OQ-02	Kategori material prioritas dan quality grade?	DLH/facilities/buyers	Before Sprint 2

ID	Question	Owner	Due
OQ-03	PJP dan model settlement yang tersedia?	Product/Legal/Finance	Before payment build
OQ-04	Apakah warga dapat menjadi buyer atau hanya verified organization?	PO/Legal	Before marketplace policy
OQ-05	Sumber knowledge internal yang boleh masuk RAG?	Knowledge Owner/Security	Before indexing
OQ-06	Integrasi pemerintah yang wajib untuk pilot?	Kominfo	Before architecture freeze
OQ-07	Baseline KPI aktual dan metode pengukuran?	Bappeda/DLH/Data Steward	Before pilot

26. TRACEABILITY MATRIX (SAMPLE)

Goal	Requirements	Journeys	Tests
Goal G-01	IAM-003, T3R-001, BS-003, OPS-001	J-01, J-03	E2E, integration
Goal G-02	MKT-001, MKT-008, PAY-001, LOG-001	J-01	Marketplace E2E
Goal G-03	AI-001..012, KN-001..007	J-04, J-05	AI eval, security
Goal G-04	T3R-002, LOG-004, ImpactLedger	J-01	Data reconciliation
Goal G-05	NFR, roadmap, UAT criteria	All	UAT, performance, DR

27. GLOSSARY

Term	Definition
Batch	Satuan material yang dapat ditelusuri berdasarkan sumber, jenis, berat, kualitas, dan status.
Diversion Rate	Persentase material yang dialihkan dari pembuangan akhir melalui reuse/recycle/recovery.
RAG	Retrieval-Augmented Generation; model mengambil bukti eksternal sebelum menghasilkan jawaban.
Groundedness	Derajat jawaban didukung oleh bukti yang diretrieval atau hasil tool terverifikasi.
PJP/PSP	Penyedia Jasa Pembayaran/Penyelenggara Jasa Sistem Pembayaran sesuai regulasi BI.

Term	Definition
RFQ	Request for Quotation; permintaan penawaran untuk spesifikasi tertentu.
Settlement	Penerusan dana kepada penerima setelah syarat transaksi terpenuhi.
Traceability	Kemampuan menelusuri asal, pergerakan, perubahan, dan tujuan material.
Impact Ledger	Catatan terstruktur tentang diversion dan dampak yang dikaitkan ke batch/order.
ABAC	Attribute-Based Access Control.
RBAC	Role-Based Access Control.

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